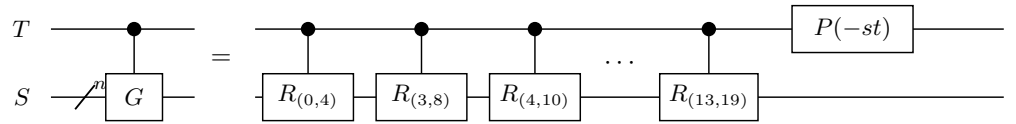
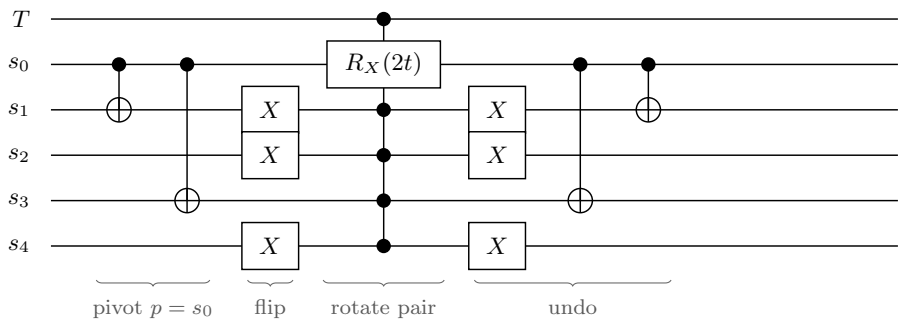
(a) Controlled evolution on the segment register, c-G (`one_bqf.py: _apply_direct_controlled_u`)



pairs in `interaction_pairs` order (upper triangle of A, row-major); worked event: $L = 6$

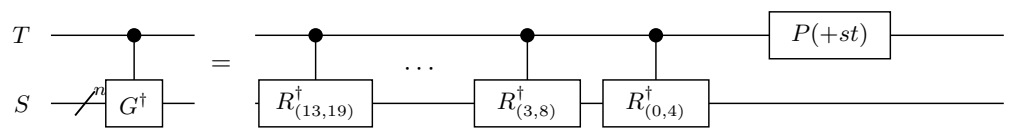
$R_{(i,j)} = e^{-it\,X_{ij}}$, $X_{ij} = |i\rangle\langle j| + |j\rangle\langle i|$
$G = e^{-ist}\prod_{\ell=1}^{L} R_{(i_\ell,j_\ell)} \approx e^{-ist}e^{-itC}$
$t = \pi/s$, so $e^{-ist} = e^{+ist} = -1$ and $G \approx e^{+iAt}$
one block per nonzero C_{ij}, $i < j$; $P(\phi) = \mathrm{diag}(1, e^{i\phi})$

(b) One pair block c-$R_{(3,8)}$ written out ($n = 5$; $|3\rangle = |00011\rangle$, $|8\rangle = |01000\rangle$, bits $s_4 \dots s_0$)



1. CNOTs from the pivot (lowest differing bit) onto the other differing bits: $|3\rangle \to |9\rangle = |01001\rangle$, $|8\rangle \to |8\rangle = |01000\rangle$; now they differ only on s_0.
2. X on every non-pivot bit that is 0 in $|9\rangle$ (s_1, s_2, s_4), so the pair's other bits read all ones.
3. $R_X(2t) = e^{-itX}$ on s_0, controlled by T and $s_1 \dots s_4$: acts only on $\mathrm{span}\{|9\rangle, |8\rangle\}$.
4. Undo 2 and 1. Net: $e^{-itX_{3,8}}$ when $T = 1$, identity otherwise.

(c) Its inverse c-$G^\dagger$, used in the uncompute (`inverse=True`)



Pairs in reverse order, each with $R_X(-2t)$ (fix 741236d on centralise-2.1). $P(\pm st)$ on the clock commutes with every block, so where it sits does not matter.